Predictive Modeling of *Salmonella enteritidis* Growth Under Sinusoidal Temperature

Modified Gompertz Model & OLS Parameter Estimation

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Introduction & Background

- *Salmonella enteritidis* is a major foodborne pathogen in egg products
- Temperature fluctuations during storage/transport affect microbial growth
- Reference: Gumudavelli et al. (2007), *J. Food Sci.* 72:M254–62 [1]

Objective

Use a modified Gompertz model to predict *Salmonella* growth under sinusoidal heating, estimate model parameters via OLS, and evaluate model adequacy.

Model Description

Primary Model (differential form)

$$\frac{d\log N}{dt} = \mu \cdot K \cdot C \cdot \exp(-K), \quad K = \exp\big(-\mu(t-M)\big) \tag{1}$$

Secondary Model (growth rate vs. temperature)

$$\mu = a(T - T_{\min})^2\big(1 - \exp[b(T - T_{\max})]\big) \tag{2}$$

- **Dependent variable**: $\log_{10} N(t)$ —log microbial concentration (log cfu/mL)
- **Independent variable**: t —time (hr)

Model Description —7 Candidate Parameters

Parameter	Symbol	Description	Units	Initial Guess
A	A	Initial log microbial number	$\log \mathrm{cfu/mL}$	$\log_{10}(400) \approx 2.60$
C	C	Upper–lower asymptote difference	$\log \mathrm{cfu/mL}$	11
M	M	Inflection point time	hr	7.5
a	a	Secondary model coefficient	$^\circ\mathrm{C}^{-2}\,\mathrm{hr}^{-1}$	0.000338
b	b	Secondary model coefficient	$^\circ\mathrm{C}^{-1}$	0.275
$T_{\min}$	$T_{\min}$	Minimum growth temperature	$^\circ\mathrm{C}$	6
$T_{\max}$	$T_{\max}$	Maximum growth temperature	$^\circ\mathrm{C}$	46.3

Data Overview

- Growth data: $\log_{10}(\text{CFU/mL})$, $n = 20$
- Temperature: sinusoidal $T(t)$ every minute
- Two independent time vectors

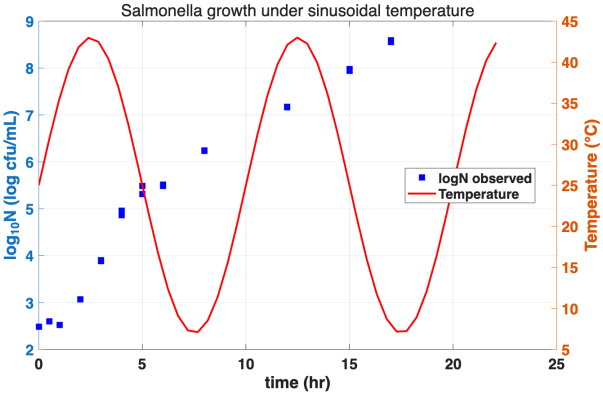

Fig 1. Dual-axis: $\log_{10} N$ and T vs. time

Forward Problem —Initial Guess

- All 7 guesses from literature
- ODE solved with `ode45`
- $T(t)$ interpolated via `interp1`

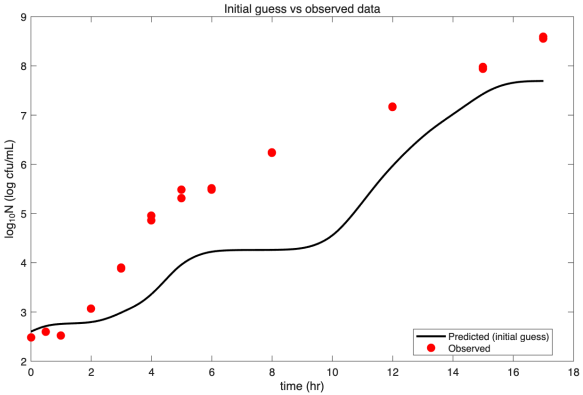

Fig 2. Initial guess vs. observed data

SSC Analysis —All 7 Parameters (Step 1)

Max|SSC|:

Param	Max|SSC|
A	2.61
C	5.11
M	5.45
a	2.20
b	**0.88**
$T_{\min}$	**0.83**
$T_{\max}$	9.02

Decision

$T_{\min}$ (0.83) and b (0.88): lowest SSC $\Rightarrow$ **fix**: $T_{\min} = 6\,^{\circ}\mathrm{C}$, $b = 0.275$.

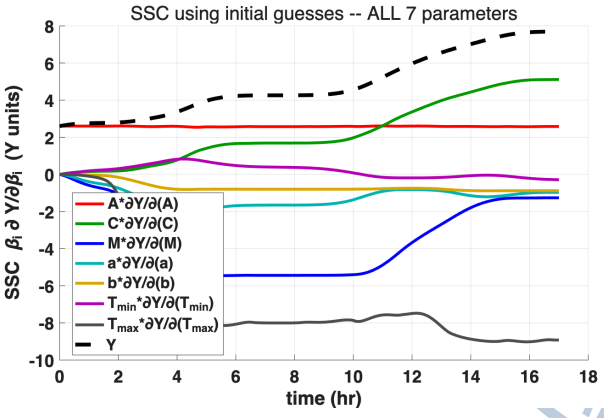

Fig 3. SSC for all 7 parameters

Parameter Selection Summary

Parameter	Max|SSC|	Identifiable?	Action	Value Used
A	2.61	✓Yes	**Estimate**	OLS result
C	5.11	✓Yes	**Estimate**	OLS result
M	5.45	✓Yes	**Estimate**	OLS result
a	2.20	✓Yes	**Estimate**	OLS result
b	0.88	× Low SSC	Fix (literature)	$0.275\,^\circ\mathrm{C}^{-1}$
$T_{\min}$	0.83	× Lowest SSC	Fix (literature)	$6\,^\circ$C
$T_{\max}$	9.02	△ C–$T_{\max}$ collinear	Fix (literature)	$46.3\,^\circ$C

- **Direct elimination**: $7 \to 4$ —fix $T_{\min}$, b (low SSC), $T_{\max}$ (C–$T_{\max}$ collinearity $R = -0.991$, fix at literature value $46.3\,^\circ$C)
- **Final estimable parameters**: $[A, C, M, a]$
- After estimation: all $|R_{ij}| < 0.60$, cond$(\mathbf{J}) = 1.19 \times 10^4 \ll 10^6$

OLS Inverse Problem —Final Results ($[A, C, M, a]$)

Param	Estimate	σ	Rel Err	95% CI
A	2.352	0.138	5.9%	[2.059, 2.645]
C	11.197	0.370	3.3%	[10.41, 11.98]
M	5.559	0.183	3.3%	[5.172, 5.946]
a	0.000300	0.0000357	11.9%	[0.0002, 0.0004]

- Fixed: $T_{\min} = 6\,^\circ$C, $b = 0.275$, $T_{\max} = 46.3\,^\circ$C (literature)
- cond$(\mathbf{J}) = 1.19 \times 10^4$ RMSE $= 0.225$ relRMSE $= 3.72\%$ $R^2 = 0.989$
- All $|R_{ij}| < 0.60$: no collinearity issues

Fitted Curve with Asymptotic CB and PB

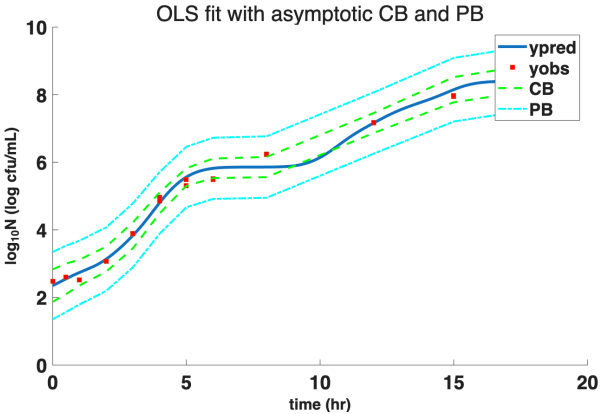

Fig 6. OLS fit with 95% simultaneous confidence bands (CB) and prediction bands (PB)

Residual Analysis

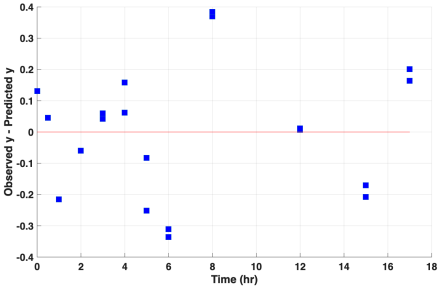

Fig 7. Residual scatter

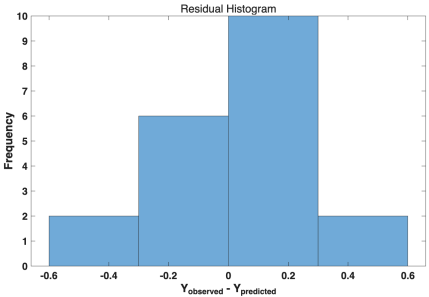

Fig 8. Residual histogram

5 Assumptions: ✓ Zero mean ($\bar{e} \approx 0.064$) △ Constant variance △ Uncorrelated (runs $= 6 < 10.5$)
✓ Normality ✓ Correct model

SSC Convergence —Before & After Estimation

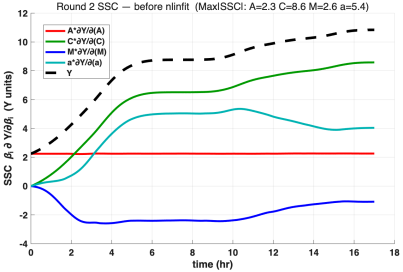

Fig 9a. SSC (initial guess)

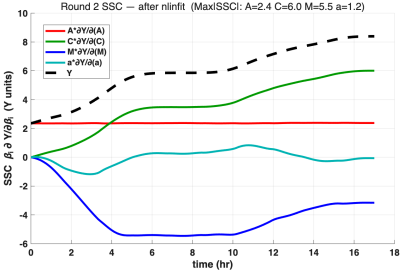

Fig 9b. SSC (after OLS)

SSC Convergence

Consistent SSC shapes before & after estimation ⇒ parameters **stable and identifiable**.

Optimal Experimental Design

- Equally-spaced `linspace` points (consistent with `inv_soln5.m`); unscaled Jacobian $J = \partial Y / \partial \beta$
- Δ criterion increases monotonically $\Rightarrow$ more data always helps
- C_{ii} (parameter variance) decreases with more data points for all 4 parameters

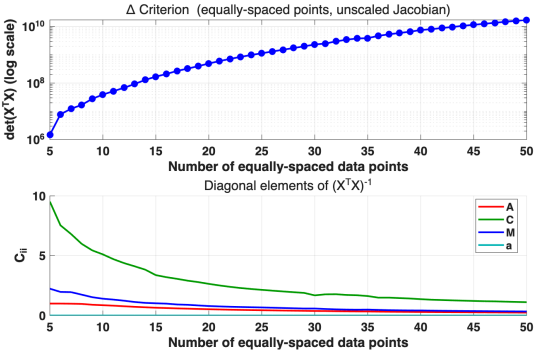

Fig 10. Δ criterion and C_{ii} vs. data points

Bootstrapping of Residuals

600 bootstrap iterations (residual resampling), 600/600 valid.

Param	Bootstrap 95% CI	Asymptotic 95% CI
A	[2.113, 2.596]	[2.059, 2.645]
C	[10.63, 12.02]	[10.41, 11.98]
M	[5.174, 5.850]	[5.172, 5.946]
a	[2.57e-4, 3.88e-4]	[2.0e-4, 4.0e-4]

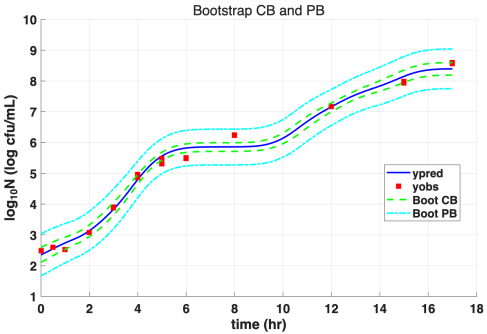

Fig 11. Bootstrap CB and PB

Asymptotic vs Bootstrap Comparison

Metric	Asymptotic	Bootstrap
CB width (avg)	0.687	**0.340**
PB width (avg)	1.858	**1.220**

- Bootstrap CB is $\mathbf{2}\times$ narrower; Bootstrap PB is $\mathbf{1.5}\times$ narrower
- Asymptotic method overestimates uncertainty for this nonlinear model
- Bootstrap provides more realistic uncertainty quantification

Conclusions

1. Modified Gompertz model successfully predicts *Salmonella* growth under sinusoidal temperature
2. **Parameter selection**: SSC analysis $\Rightarrow$ fix $T_{\min}$, b (low SSC) and $T_{\max}$ (C–$T_{\max}$ collinearity $R = -0.991$); estimate $[A, C, M, a]$ ($7 \to 4$ direct elimination)
3. Robust OLS fit: cond($\mathbf{J}$) $= 1.19 \times 10^4$, relRMSE $= 3.72\%$, $R^2 = 0.989$; all $|R_{ij}| < 0.60$; residual autocorrelation (runs $= 6 < 10.5$) suggests model improvement potential
4. Bootstrap CI narrower than asymptotic CI ($2\times$ for CB, $1.5\times$ for PB) $\Rightarrow$ asymptotic method overestimates uncertainty

References I

[1] Lavanya Gumudavelli, Prem Rishi Velugoti, Ramaswamy C Anantheswaran, John D Floros, and Stephen J Knabel.
Dynamic predictive model for growth of *Salmonella Enteritidis* in egg yolk.
Journal of Food Science, 72(7):M254–M262, 2007.